# Diet and risk of diverticular disease in Oxford cohort of European Prospective Investigation into Cancer and Nutrition (EPIC): prospective study of British vegetarians and non-vegetarians

Objective To examine the associations of a vegetarian diet and dietary fibre intake with risk of diverticular disease. Design Prospective cohort study. Setting The EPIC-Oxford study, a cohort of mainly health conscious participants recruited from around the United Kingdom. Participants 47 033 men and women living in England or Scotland of whom 15 459 (33%) reported consuming a vegetarian diet. Main outcome measures Diet group was assessed at baseline; intake of dietary fibre was estimated from a 130 item validated food frequency questionnaire. Cases of diverticular disease were identified through linkage with hospital records and death certificates. Hazard ratios and 95% confidence intervals for the risk of diverticular disease by diet group and fifths of intake of dietary fibre were estimated with multivariate Cox proportional hazards regression models. Results After a mean follow-up time of 11.6 years, there were 812 cases of diverticular disease (806 admissions to hospital and six deaths). After adjustment for confounding variables, vegetarians had a 31% lower risk (relative risk 0.69, 95% confidence interval 0.55 to 0.86) of diverticular disease compared with meat eaters. The cumulative probability of admission to hospital or death from diverticular disease between the ages of 50 and 70 for meat eaters was 4.4% compared with 3.0% for vegetarians. There was also an inverse association with dietary fibre intake; participants in the highest fifth (≥25.5 g/day for women and ≥26.1 g/day for men) had a 41% lower risk (0.59, 0.46 to 0.78; P<0.001 trend) compared with those in the lowest fifth (<14 g/day for both women and men). After mutual adjustment, both a vegetarian diet and a higher intake of fibre were significantly associated with a lower risk of diverticular disease. Conclusions Consuming a vegetarian diet and a high intake of dietary fibre were both associated with a lower risk of admission to hospital or death from diverticular disease.